



Project Context

This project simulates a Revenue Growth Management (RGM) analytics workflow inspired by a real-world tobacco business.

The original dataset used was the Superstore dataset, transformed into a tobacco-style commercial dataset. Key constructs introduced:

- Oral Nicotine, Cigarettes, Heated Tobacco
- List price vs Net invoice price
- Trade spend (discounts & incentives)
- Weekly commercial performance tracking

 *The purpose of this stage is decision-oriented commercial analytics, not machine learning.*

Objective of This Stage

This stage answers one core business question:

“Given prices, volumes, and trade spend – where should the business act?”

To answer this, the focus was on:

- KPI construction
- Trade effectiveness analysis
- Price realization & leakage detection
- Segment-level diagnostics
- Creation of a Decision Pack

Data Grain

Weekly (aligned with RGM operating cadence)

AGGREGATIONS PERFORMED AT:

Country

Channel

PMI Product Family

Brand Tier

Wholesaler Tier

What Was Done

Analytical Approach & Methodology



KPI Dataset Construction

01

- Gross Revenue
- Trade Spend
- Net Revenue (Post-Trade)
- Shipment Units
- Average List Price
- Average Net Price
- Price Realization %
- Margin Proxy

Why this matters:

RGM teams require a single source of truth before moving to diagnostics or dashboards.



Price-Volume Relationship

02

Correlation analyzed between:

-  Avg Net Price
-  Shipment Units

Result: Most segments show negative correlation (realistic elasticity).

Interpretation: Higher prices generally reduce volume — dataset behaves like real market.



Trade Effectiveness Analysis

03

Analyzed **Trade Spend** vs **Net Revenue After Trade** at Country–Channel level.

Result: Some segments show strong revenue response, others show weak/inefficient response.

Interpretation: Trade spend is not uniformly productive and requires targeting.



Price Realization & Leakage Analysis

04

Key Metrics Calculated:

- **Price Realization %:** Percentage of list price captured as net revenue.
- **Leakage per Unit:** List Price – Net Price

Result: Overall realization $\approx$ 84%, high leakage in certain segments.

Interpretation: Discounting discipline breaks down in specific segments.



4 Price Realization (Continued)

Result:

- Overall price realization $\approx$ 84%
- Certain segments show high leakage per unit

Interpretation:

Discounting discipline breaks down in specific segments, indicating optimization opportunities.

5 Rolling Weekly Trends

4-week rolling trends were created for:

- Gross Revenue
- Trade Spend %
- Price Realization

Why this matters: RGM teams manage trends, not single weeks. Rolling views smooth volatility and show direction.

What Is a Decision Pack?

A Decision Pack is a management-ready summary, not a technical artifact.

It is designed to answer:

- ✓ Are we making money?

✓ Is trade spend under control?
- ✓ Is pricing holding?

✓ Where should we intervene?

Key Decision Pack Results

Aggregated performance metrics from the analysis

Metric	Value
Total Gross Revenue	~\$12.6M
Total Trade Spend	~\$1.35M
Trade Spend % of Revenue	~10.6%
Net Revenue After Trade	~\$11.3M
Avg Price Realization	~84%

Business Meaning

- ▶ **Trade spend** is meaningful but not excessive.
- ▶ **Price leakage** exists and is segment-specific, indicating inefficiencies.
- ▶ Clear **prioritization opportunities** exist for RGM actions to capture more value.



Strategic Insights Enabled

This analysis enables decisions such as:

- ✓ **Reducing trade spend** in low-return segments to eliminate waste.
- ✓ **Protecting price** in high-realization segments where brand power is strong.
- ✓ **Redesigning incentives** where leakage is highest to improve margin capture.
- ✓ **Prioritizing** price-sensitive markets and channels for targeted interventions.



Why This Stage Stops Here

No predictive modeling was done intentionally.

RGM teams first require:

- ✓ Clean KPIs
- ✓ Transparent diagnostics
- ✓ Confidence in numbers

→ Only then do dashboards and strategy discussions begin.

Next Step: Power BI (DAX & Dashboards)

The next phase will:

 Import KPI and decision tables into Power BI

Create DAX for:

- ▶ Trade efficiency
- ▶ Price realization
- ▶ Leakage monitoring

Build:

- ▶ Executive KPI dashboard
- ▶ Pricing diagnostics view
- ▶ Trade optimization page

 This completes the flow: SQL → Analytics → Decisions → Visualization



This analytics stage has successfully transformed raw **transactional data** into meaningful business assets.



Trustworthy KPIs

Single source of truth established through rigorous data cleaning and calculation.



Clear Commercial Signals

Uncovered price leakage, trade inefficiency, and volume elasticity trends.



Decision-Ready Foundation

Structured outputs prepared specifically for management review and action.